

Comprehensive Local Development Guide

ICAR Project (Drupal 11 & Docker)

This document contains the complete, official blueprint and step-by-step instructions for setting up, running, and developing the ICAR Project using Drupal 11 and Docker.

Part 1: The Blueprint Files (Repository Foundation)

These are the core configuration files that should be committed to your Git repository. They ensure every team member has the exact same server environment.

1. **Dockerfile**

```
FROM php:8.4-apache

# Install system dependencies and PHP extensions required for Drupal 11
RUN apt-get update && apt-get install -y      libpng-dev      libjpeg-dev      libpq-dev
libxml2-dev      libzip-dev      zip      unzip      git      && rm -rf /var/lib/apt/lists/*

# Configure and install PHP extensions
RUN docker-php-ext-configure gd --with-jpeg      && docker-php-ext-install -j$(nproc)
gd      opcache      pdo      pdo_mysql      zip

# Configure recommended PHP.ini settings for Drupal
RUN {      echo 'opcache.memory_consumption=128';      echo
'opcache.interned_strings_buffer=8';      echo 'opcache.max_accelerated_files=4000';
echo 'opcache.revalidate_freq=60';      echo 'opcache.fast_shutdown=1';      echo
'opcache.enable_cli=1';      } > /usr/local/etc/php/conf.d/opcache-recommended.ini

# Enable Apache rewrite module for Drupal clean URLs
RUN a2enmod rewrite

# Install Composer globally
COPY --from=composer:latest /usr/bin/composer /usr/bin/composer

# Set the working directory
WORKDIR /var/www/html

# Adjust Apache configuration to point to the 'web' subdirectory
RUN sed -i 's|/var/www/html|/var/www/html/web|g' /etc/apache2/sites-available/000-
default.conf
```

2. `docker-compose.yml`

```
services:
  drupal:
    build: .
    container_name: drupal_app
    ports:
      - "8080:80"
    volumes:
      - ./var/www/html
    environment:
      - DRUPAL_DATABASE_HOST=db
      - DRUPAL_DATABASE_NAME=drupal_db
      - DRUPAL_DATABASE_USER=drupal_user
      - DRUPAL_DATABASE_PASSWORD=drupal_password
    depends_on:
      - db

  db:
    image: mariadb:10.11
    container_name: drupal_db
    restart: always
    environment:
      MYSQL_DATABASE: drupal_db
      MYSQL_USER: drupal_user
      MYSQL_PASSWORD: drupal_password
      MYSQL_ROOT_PASSWORD: root_password
    volumes:
      - db_data:/var/lib/mysql
    ports:
      - "3307:3306"

volumes:
  db_data:
```

3. `.gitignore`

```
# Ignore Drupal Core & Vendor files (Composer handles these)
vendor/
web/core/
web/modules/contrib/
web/themes/contrib/
web/profiles/contrib/

# Ignore user uploads and generated files
web/sites/default/files/
web/sites/default/settings.php

# Ignore environment files
.env
```

Part 2: Team Member Setup Guide

Follow these steps to get the project running on your local machine. Ensure you have Git, Docker Desktop, and WSL (if on Windows) installed.

Step 1: Clone the Repository

```
git clone https://github.com/SamarthThombre/ICAR-Project.git
cd ICAR-Project
```

Step 2: Build the Docker Environment

```
sudo docker compose up -d --build
```

Step 3: Install Drupal Dependencies

```
sudo docker compose exec drupal composer install
```

(Note: If you get a "Dubious Ownership" Git error here, see the Troubleshooting section below).

Step 4: Configure Installer Permissions

The web server needs permission to create settings files and upload directories. Run these exactly as written:

```
sudo docker compose exec drupal cp web/sites/default/default.settings.php web/sites/default/settings.php
sudo docker compose exec drupal mkdir -p web/sites/default/files
sudo docker compose exec drupal chown -R www-data:www-data web/sites/default
sudo docker compose exec drupal chmod -R 775 web/sites/default
```

Step 5: Browser Installation

1. Go to **http://localhost:8080**
2. Choose your language and select "Standard" installation.
3. On the Database Configuration screen, click **Advanced Options**.
4. Change Host from **localhost** to **db**.
5. Enter credentials:
 - Database name: **drupal_db**
 - Database username: **drupal_user**
 - Database password: **drupal_password**
6. Complete the setup with your preferred admin details.

Part 3: Troubleshooting

Git "Dubious Ownership" Error during Composer Install

If Step 3 fails with `fatal: detected dubious ownership in repository at '/var/www/html'`, run this command to add a security exception inside the container:

```
sudo docker compose exec drupal git config --global --add safe.directory /var/www/html
```

Then run the composer install command again:

```
sudo docker compose exec drupal composer install
```

Apache "Service is not running" Error

If your container fails to start because the `web` folder doesn't exist yet (usually only happens on the very first brand-new build before composer downloads Drupal), use a temporary container to download the files:

```
sudo docker compose run --rm drupal composer create-project drupal/recommended-project temporary-dir --no-interaction
sudo docker compose run --rm drupal bash -c "mv temporary-dir/* temporary-dir/. * 2>/dev/null || true && rmdir temporary-dir"
sudo docker compose up -d
```